



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO AMMAN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

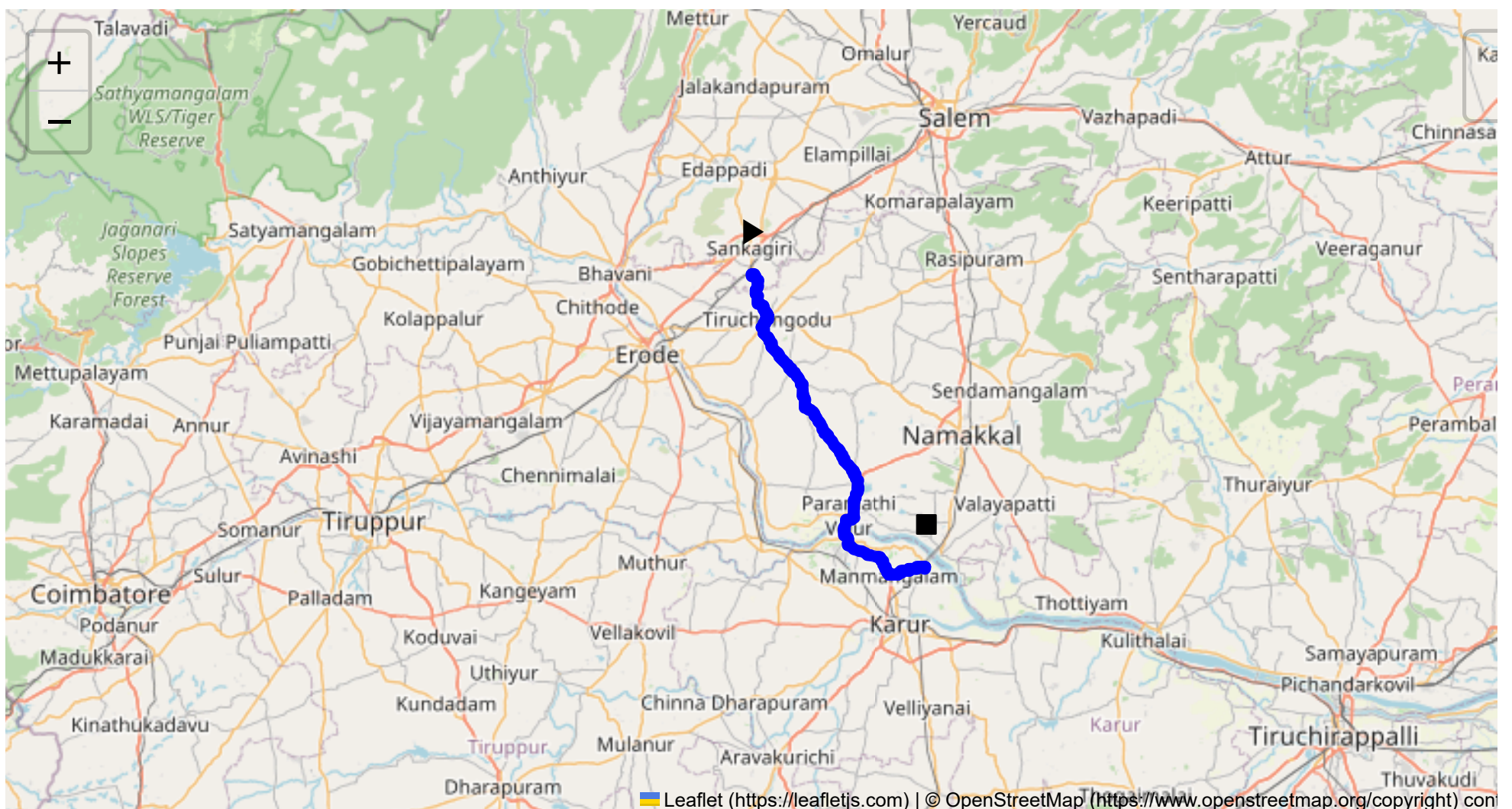
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 63.12 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours
Start: (11.4381, 77.8734)
End: (11.036488, 78.115654)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 24P8+G49, Vangal spans approximately 63.12 kilometers. The journey typically moves through several towns and highways, including Puthur, Thondikaradu, Velur, and Manmangalam, using roads like the Sankagiri-Tiruchengode Road and the Karur-Namakkal Highway. This path is relatively direct, weaving through both urban and rural settings.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate, with hot and dry weather dominating. During the summer months, temperatures can soar, leading to the possibility of road surface damage or tire blowouts. The monsoon season, from June to September, often brings heavy rains, which may result in flooding and slippery roads, particularly on less maintained rural roads.

3. Traffic Patterns and Congestion-Prone Areas

- **Peak Hours:** Typically, congestion is noted in the mornings (8:00 AM - 10:00 AM) and evenings (5:00 PM - 7:00 PM), especially near towns like Tiruchengode, Velur, and Manmangalam.
- **Congestion Areas:** Major choke points include towns with narrow roads and high pedestrian traffic, notably near schools such as the Girls High School in Thondikaradu.

4. Assessment of Road Quality and Infrastructure

- **Urban Areas:** Roads are generally well-paved in urban settings, though some parts may suffer from potholes due to heavy traffic and poor drainage during monsoons.
- **Rural Areas:** Some rural segments may have uneven surfaces and require caution, particularly during adverse weather conditions.

5. Suggestions for Alternative Routes for Emergencies

- **Bypass Roads:** In emergencies such as roadblocks or accidents, using bypass roads before reaching urban congestion points in towns like Tiruchengode could help.
- **Alternative Highways:** The NH-44 could serve as an alternative for long stretches but may require additional detours to rejoin the main path toward Vangal.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits Required:** Transporting hazardous materials requires permits from local transport authorities.
- **Time Restrictions:** Movement might be restricted during peak hours in urban areas for safety.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

This route has occasionally recorded incidents involving vehicular breakdowns, particularly in monsoon-heavy months where waterlogging can lead to accidents. There are no significant historical reports of major hazardous material incidents, but preventive caution is advised.

8. Environmental Considerations and Sensitive Areas

- **Near Schools:** Sections near schools, such as in Thondikaradu, are sensitive due to high pedestrian activities.
- **Agricultural Zones:** Parts of the route run through agricultural zones; minimizing pollution and disruptions is crucial.

9. Analysis of Communication Coverage

- **Urban Coverage:** Generally strong communication coverage in towns.
- **Rural Dead Zones:** Sporadic weak connectivity between Puthur and Velur. Backup communication devices are recommended.

10. Estimated Emergency Response Times for Different Route Segments

- **Urban Areas:** Emergency response (ambulance, police) is typically swift, within 15-30 minutes.
- **Rural Areas:** Response times may extend to 30-60 minutes due to lesser infrastructure and resources.

12. Overall Summary of Risk Assessment

The route presents typical risks associated with transportation through a mix of urban and rural environments. Key risk factors include weather-induced hazards during monsoons, congestion in urban towns, and variable road quality in rural sectors. However, with adherence to local regulations, proper preparation for adverse weather, and careful navigation of congested zones, the route can be effectively managed for heavy vehicles, including those carrying hazardous materials. Continuous monitoring of weather forecasts and traffic conditions, along with ensuring effective communication tools, will further mitigate risks throughout the journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43961, 77.87341	15 KM/Hr
3	Turn	High	11.44022, 77.87546	15 KM/Hr
4	Turn	High	11.37862, 77.89488	15 KM/Hr
5	Turn	High	11.37850, 77.89453	15 KM/Hr
6	Turn	Medium	11.16749, 78.01572	30 KM/Hr
7	Turn	Medium	11.15573, 78.02176	30 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Blind Spot	Blind Spot	11.14480, 78.02291	10 KM/Hr
10	Turn	High	11.02679, 78.06555	15 KM/Hr
11	Turn	Medium	11.02740, 78.07688	30 KM/Hr
12	Turn	Medium	11.03850, 78.11244	30 KM/Hr
13	Blind Spot	Blind Spot	11.03771, 78.11595	10 KM/Hr
14	Turn	High	11.03684, 78.11528	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
6	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44022, 77.87546



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16749, 78.01572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14545, 78.02208



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.02679, 78.06555



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.02740, 78.07688



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.03771, 78.11595



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.03684, 78.11528
